

Announcements

- HW3 is out
- Mid semester Anonymous Survey Out on Canvas. Due this Sunday
- Approx Mean of Midterm is 75-76% for this section

Recap

- PCA used for?
- PCA supervised or unsupervised?

dimensionality reduction.

$$X = \begin{bmatrix} \end{bmatrix}$$

$$Y = \begin{bmatrix} \text{target} \\ X \end{bmatrix}$$

x space $\longrightarrow$ z space
 d d

Supervised dimensionality reduction - Forward feature Selection

- Start with **no features**.
- Add features **one at a time** — at each step, choose the feature that gives the biggest improvement to model performance (e.g., accuracy, R^2 , AIC).
- Stop when adding new features no longer improves performance beyond threshold.

$$X = \begin{bmatrix} \text{ht} \cdot \text{wt} \cdot \text{e} \cdot \text{f} \end{bmatrix}$$

model $\rightarrow$ 92% accuracy

$$\begin{bmatrix} \text{ht} \end{bmatrix}$$

$$\begin{bmatrix} \text{wt} \end{bmatrix}$$

$$\begin{bmatrix} \text{e} \end{bmatrix}$$

$$\begin{bmatrix} \text{f} \end{bmatrix}$$

$\rightarrow$ NB $\rightarrow$ 82%

70%

50%

35%

Best (2) features
Reduce dimensions
Conserving 90%
of original accuracy

$$\underline{0.9 \times 0.92}$$

Backward Feature Selection

- Start with **all features**.
- Remove features **one at a time** — at each step, drop the feature whose removal least reduces model performance.
- Stop when removing any more features harms performance too much.

h w e f h w e h e f h w f

[

Both forward and backward feature selection are quite computationally expensive

more computationally expensive

Linear Regression

Nimisha Roy
Georgia Tech

The Moment

© Yongqing Bao

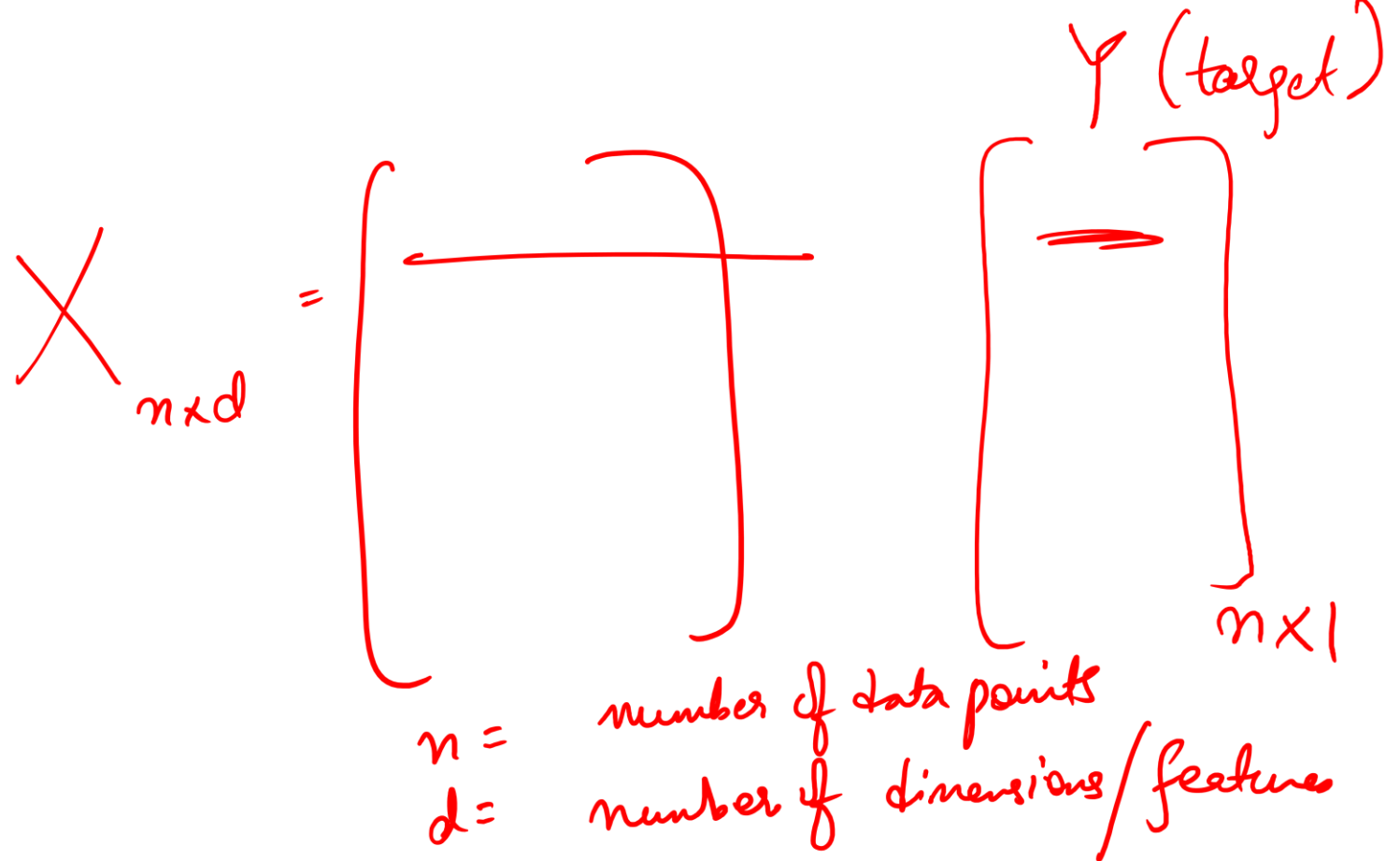


Incoming ML HW3

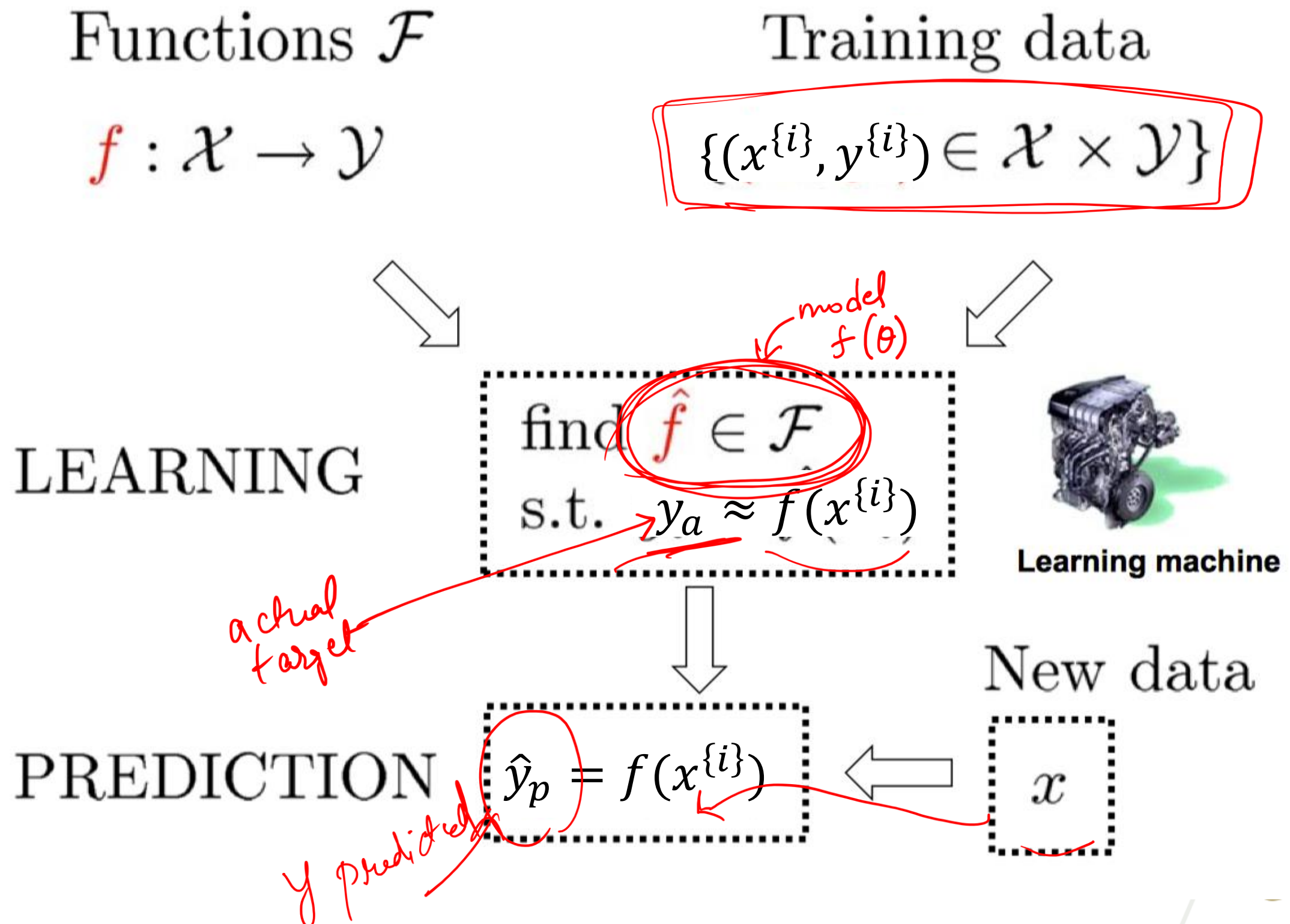
You; after HW2

Outline

- Supervised Learning
- Linear Regression
- Extension



Supervised Learning: Overview



Supervised Learning: Split of data

Training Set

70% of dataset

- **Purpose:** Used to **fit** the model — the algorithm learns patterns, relationships, and weights from this data.
- **What happens here:** The model adjusts its internal parameters to minimize error.
- **Analogy:** Like studying the material before an exam.

Validation Set

10%

- **Purpose:** Used to **tune hyperparameters** and make design decisions (e.g., how many layers, what learning rate, etc.).
- **What happens here:** The model is not trained on this data; instead, it helps to evaluate how changes affect performance, preventing overfitting.
- **Analogy:** Like taking practice tests to decide which study strategy works best.

Test Set

20%

- **Purpose:** Used to **assess the final performance** of the model after all training and tuning.
- **What happens here:** No further learning or tweaking — it provides an **unbiased estimate** of the model's generalization ability.
- **Analogy:** Like the final exam — measures how well you really learned the material.

Supervised Learning: Two Types of Tasks

Given: training data

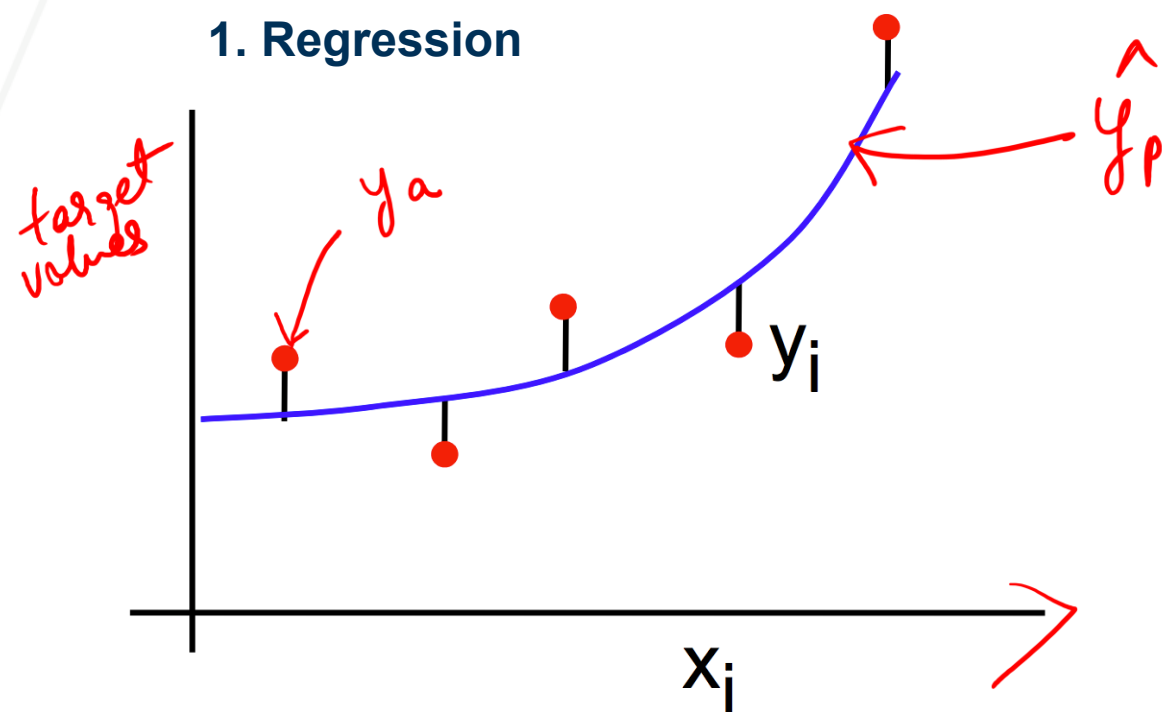
$$\{(x^{\{1\}}, y^{\{1\}}), (x^{\{2\}}, y^{\{2\}}), \dots, (x^{\{n\}}, y^{\{n\}})\}$$

Learn: a function

$$f(\mathbf{x}) : y = f(\mathbf{x})$$

When y is continuous:

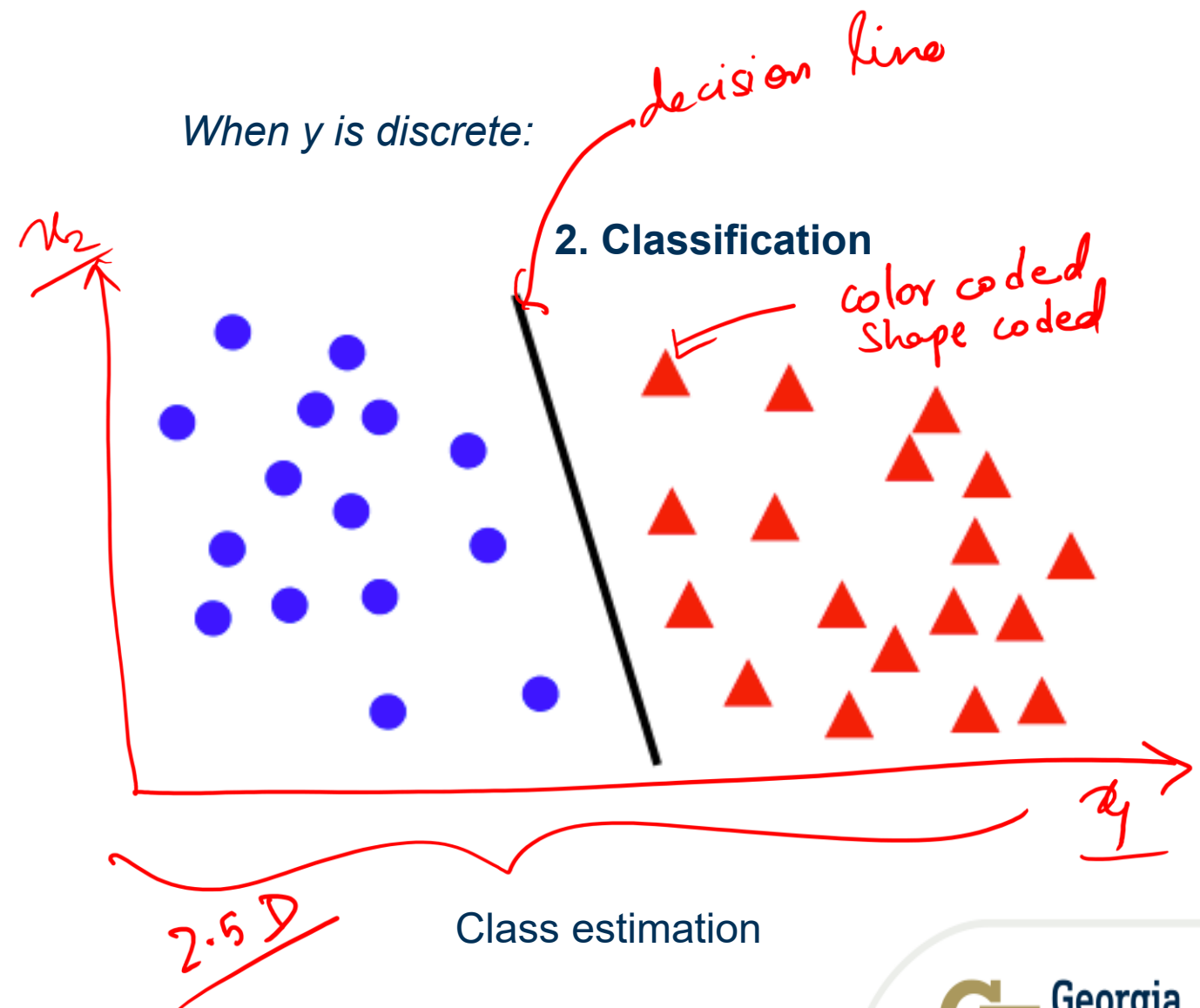
1. Regression



Curve fitting

When y is discrete:

2. Classification

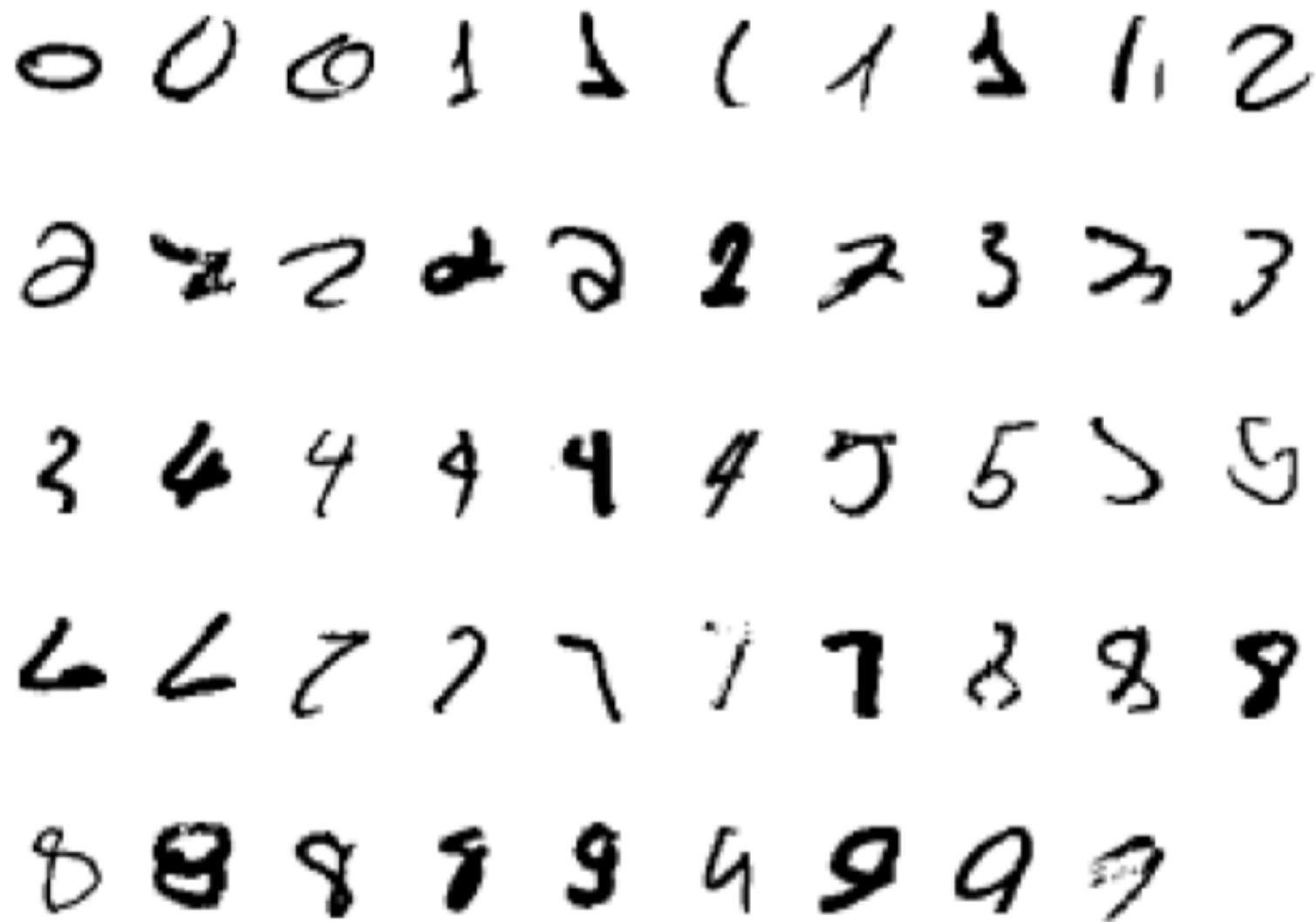


Classification Example 1: Handwritten digit recognition

As a supervised classification problem

10 classes

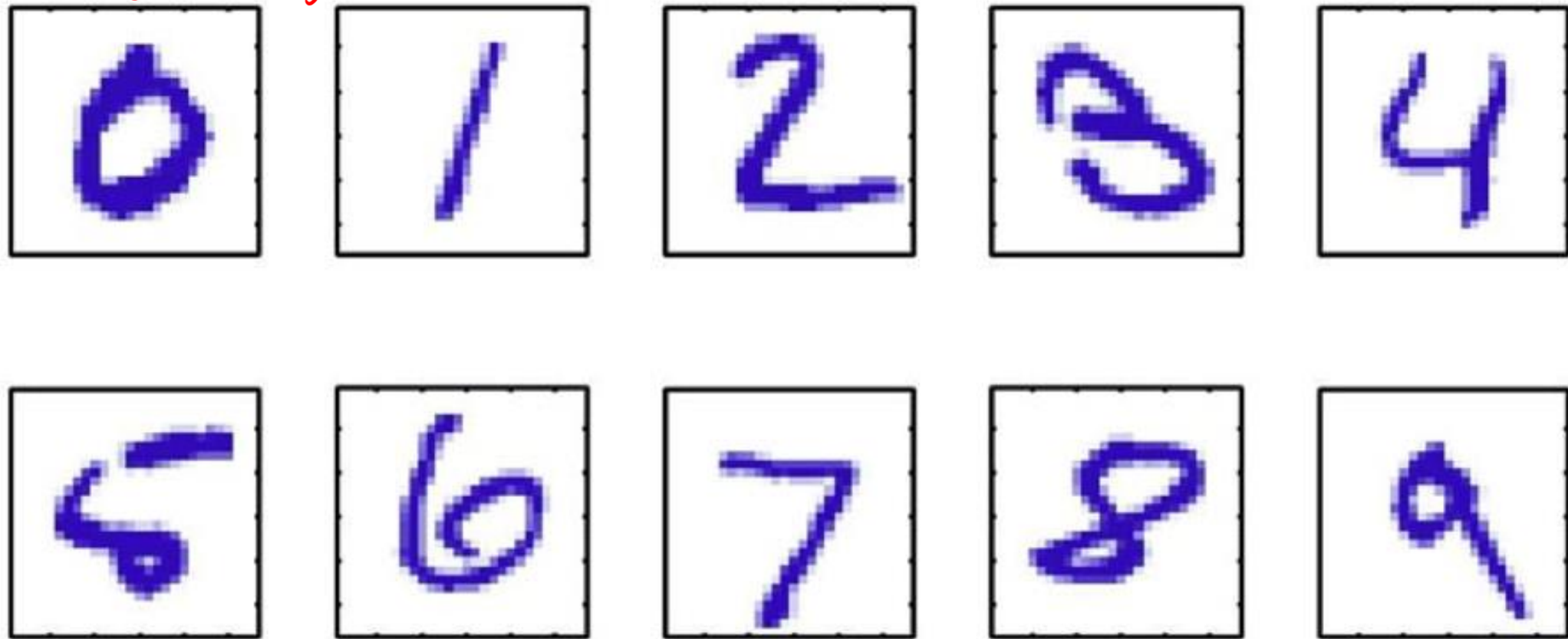
Start with training data, e.g. 6000 examples of each digit



- Can achieve testing error of 0.4%
- One of first commercial and widely used ML systems (for zip codes & checks)

Classification Example 1: Hand-Written Digit Recognition

*dimensionality
reducing using
forward or backward
feature selection*



Images are 28 x 28 pixels

A classification problem

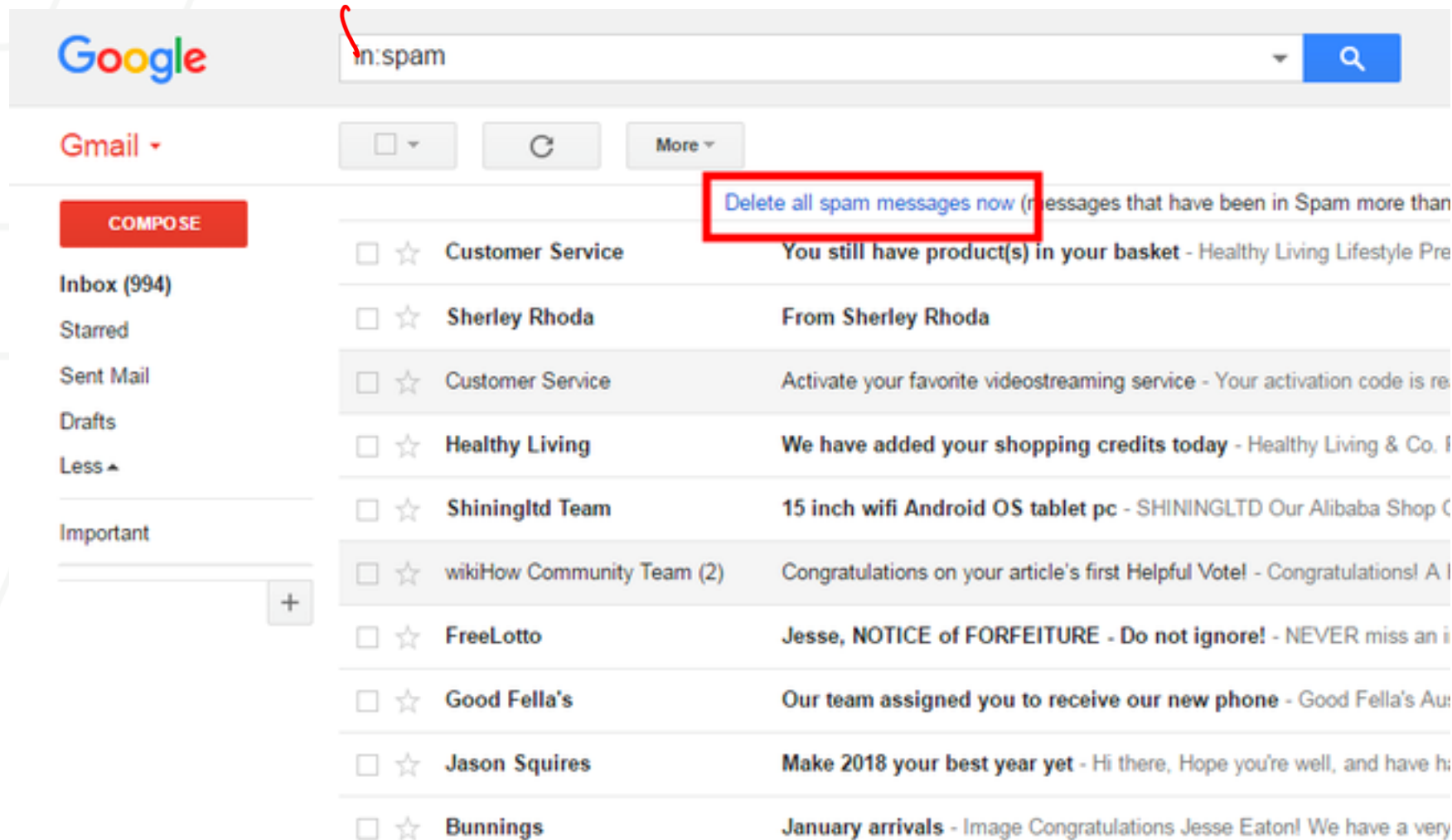
dimension $\rightarrow 784$

Represent input image as a vector $\mathbf{x} \in \mathbb{R}^{784}$

Learn a classifier $f(\mathbf{x})$ such that,

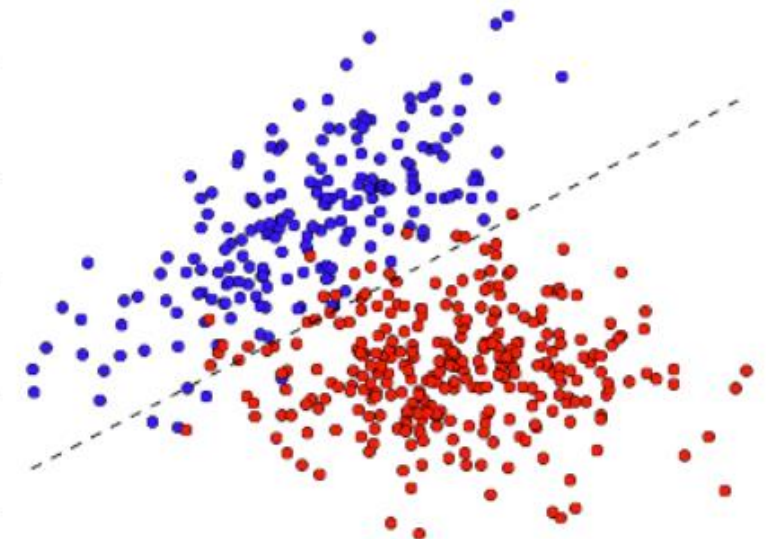
$$f : \mathbf{x} \rightarrow \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

Classification Example 2: Spam Detection



① Vocabulary/
Dictionary

NOT SPAM



SPAM

A classification problem

- This is a classification problem
- Task is to classify email into spam/non-spam
- Data x_i is word count
- Requires a learning system as “enemy” keeps innovating

Regression Example 1: Apartment Rent Prediction

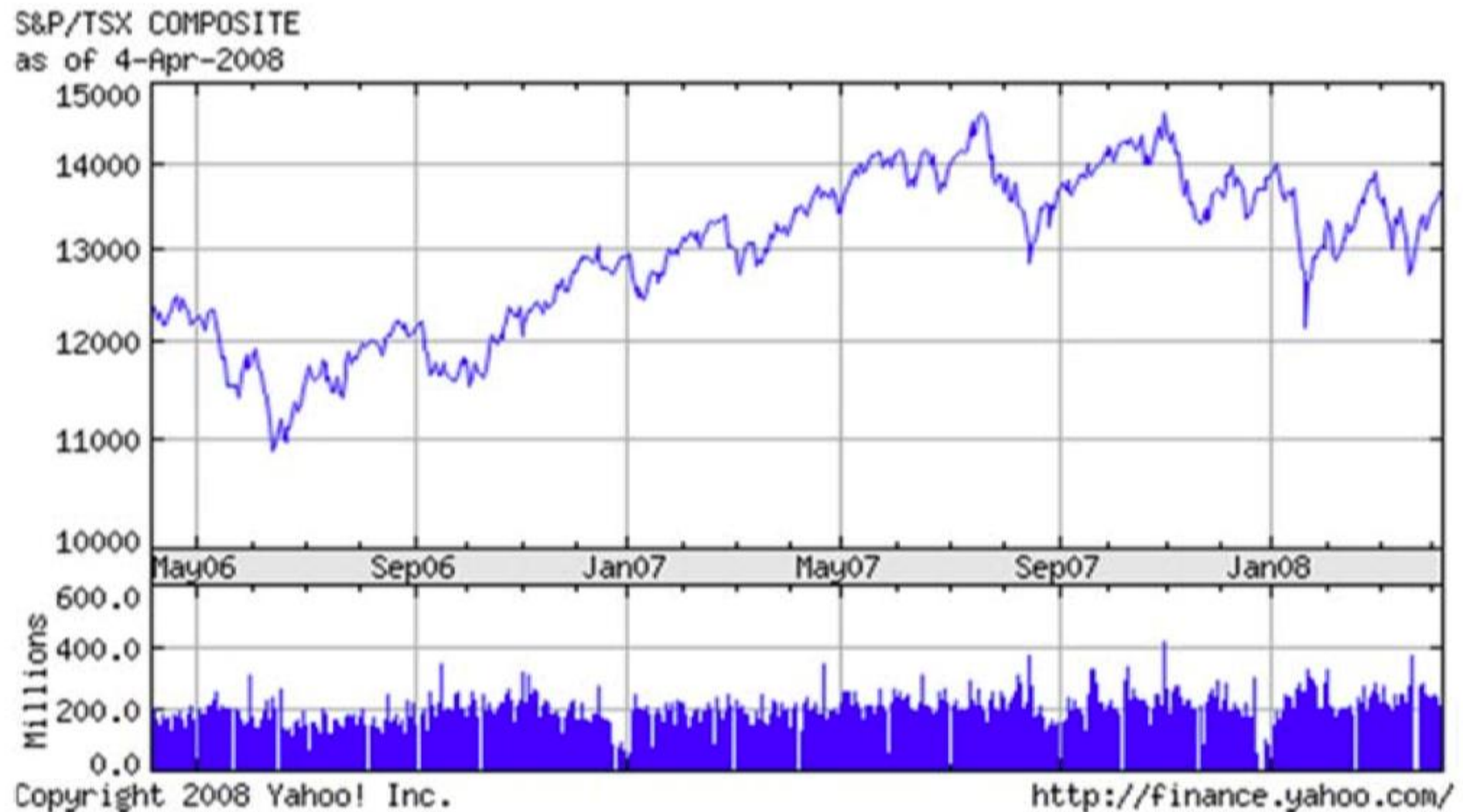
- Suppose you are to move to Atlanta
- And you want to find the **most reasonably priced** apartment satisfying your **needs**:

square-ft., # of bedroom, distance to campus ...

A regression problem

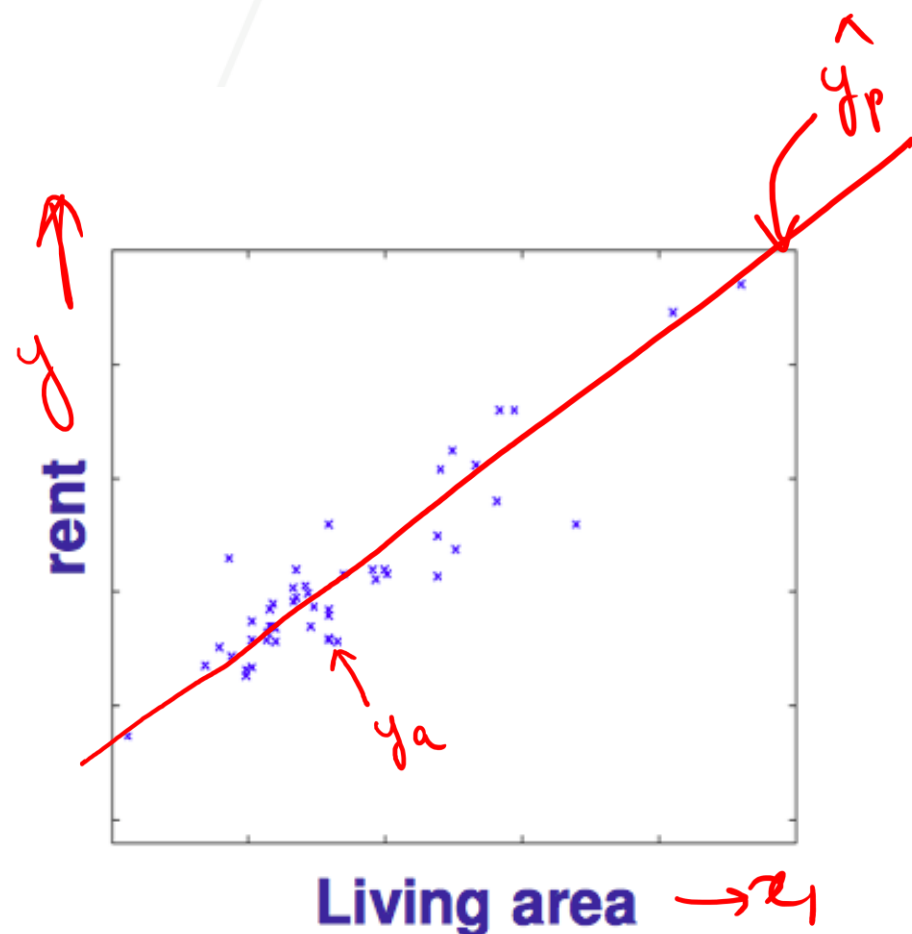
Living area (ft ²)	# bedroom	Rent (\$)
230	1	600
506	2	1000
433	2	1100
109	1	500
...		
150	1	?
270	1.5	?

Regression Example 2: Stock Price Prediction



- Task is to predict stock price at future date

A regression problem



- Features:

- Living area, distance to campus, # bedroom ...
- Denote as $x = (x_1, x_2, \dots, x_d)$

- Target:

- Rent
- Denoted as y

$$\hat{y}_p = mx + b \leftarrow \text{offset or } y\text{-intercept}$$

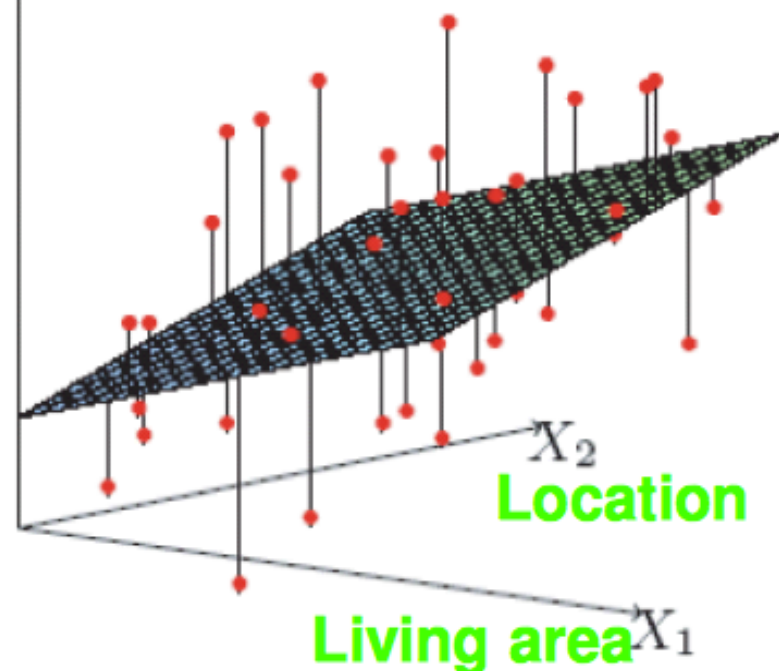
notation

$$\hat{y}_p = \theta_0 + \boxed{\theta_1 x_1}$$

bias term

gateways to features

rent



Training set:

- $x = \{x^{(1)}, x^{(2)}, \dots, x^{(n)}\} \in R^d$
- $y = \{y^{(1)}, y^{(2)}, \dots, y^{(n)}\}$

$$\hat{y}_p = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

d features $\rightarrow (d+1)$ parameters

$$x^{(i)} = [1 \quad x_1 \quad x_2 \quad x_3 \dots x_d]$$

(1 x d+1)

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix}_{(d+1) \times 1}$$

$$y = \theta_0^2 + \theta_1^2 x_1 + \dots + \theta_d^2 x_d$$

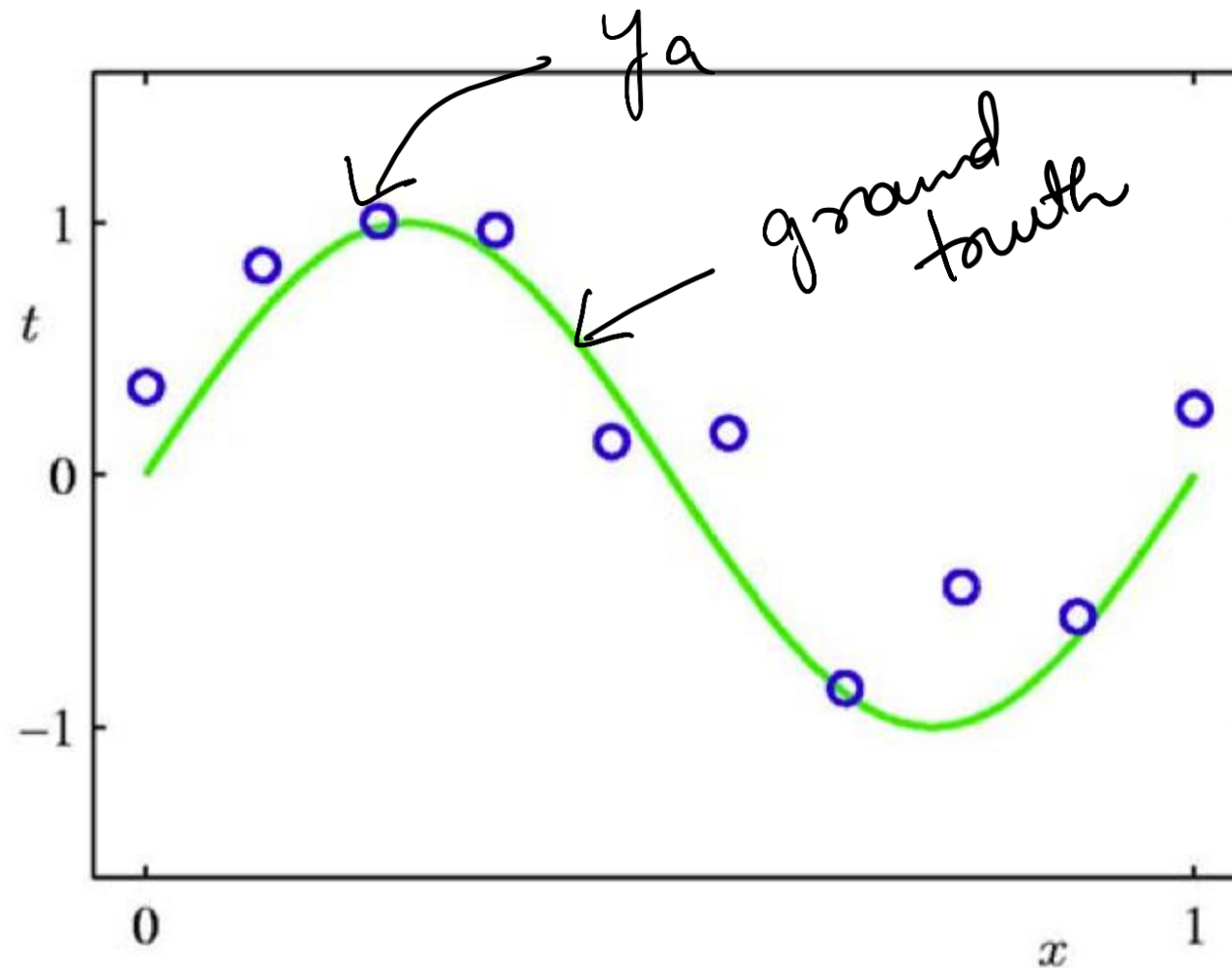
X LCF

$$y = \theta \cdot x$$

LCF $\leftarrow$ Linear Combination of Features

$$y = \theta_0 + \theta_1 \underbrace{x_1^2}_{\text{LCF}} + \theta_2 x_2^2 + \dots + \theta_d x_d^2$$

Regression: Problem Setup



- Suppose we are given a training set of N observations
- Regression problem is to estimate $y(x)$ from this data

Outline

- Supervised Learning
- Linear Regression ←
- Extension

Linear Regression

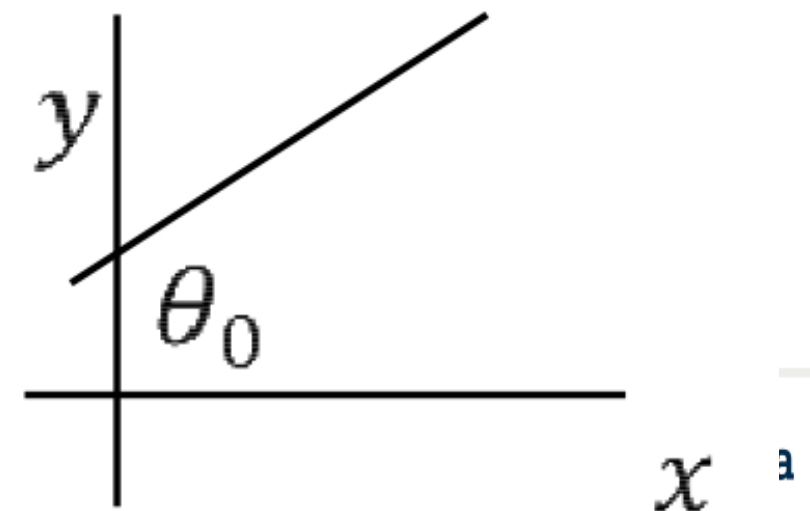
- Assume y is a linear function of x (features) plus noise ϵ

$$y = \theta_0 + \theta_1 x_1 + \dots + \theta_d x_d + \epsilon$$

← noise

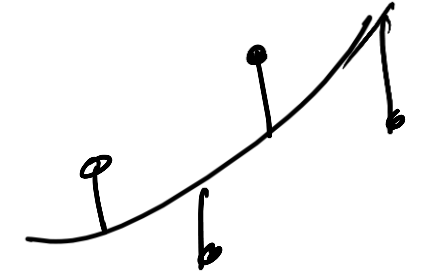
- where ϵ is an error term of unmodeled effects or random noise
- Let $\theta = (\theta_0, \theta_1, \dots, \theta_d)^\top$, and augment data by one dimension

- Then $y = x\theta + \epsilon$



Least Mean Square Method

$$\|y^{(i)} - x^{(i)}\theta\| E[\theta]$$



- Given n data points, find θ that minimizes the mean square error

Training

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \underbrace{L(\theta)}_{E(\theta)} = \frac{1}{n} \sum_{i=1}^N \underbrace{\left(\underbrace{y^{(i)}}_{y_a} - \underbrace{x^{(i)}\theta}_{\hat{y}_p} \right)^2}_{1 \times 1 \quad x^{(d+1)} \quad (d+1) \times 1}$$

- Our usual trick: set gradient to 0 and find parameter

$$\frac{\partial L(\theta)}{\partial \theta} = 0$$

$$\frac{\partial L(\theta)}{\partial \theta} = -\frac{2}{n} \sum_{i=1}^N (x^{(i)})^T (y^{(i)} - x^{(i)}\theta) = 0$$

$$\frac{\partial L(\theta)}{\partial \theta} = -\frac{2}{n} \underbrace{\sum_{i=1}^N (x^{(i)})^T y^{(i)}}_{\text{}} + \frac{2}{n} \sum_{i=1}^N (x^{(i)})^T x^{(i)} \theta = 0$$

Matrix form

$$x = \begin{bmatrix} 1 & x_1^{\{1\}} & \dots & x_d^{\{1\}} \\ 1 & x_1^{\{2\}} & \ddots & x_d^{\{2\}} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^{\{n\}} & \dots & x_d^{\{n\}} \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix}$$

$n \times (d + 1) \qquad n \times 1 \qquad (d + 1) \times 1$

$$MSE(\theta) = \underset{\theta}{\operatorname{argmin}} L(\theta) = \frac{1}{n} \underbrace{(y - x\theta)^T} \underbrace{(y - x\theta)}$$

$$x\theta = \begin{bmatrix} \theta_0 + \theta_1 x_1^{\{1\}} + \theta_2 x_2^{\{1\}} + \dots + \theta_d x_d^{\{1\}} \\ \theta_0 + \theta_1 x_1^{\{2\}} + \theta_2 x_2^{\{2\}} + \dots + \theta_d x_d^{\{2\}} \\ \vdots \\ \theta_0 + \theta_1 x_1^{\{n\}} + \theta_2 x_2^{\{n\}} + \dots + \theta_d x_d^{\{n\}} \end{bmatrix}$$

$n \times 1$

Matrix Version and Optimization

$$y = X\theta.$$
$$\theta = \bar{X}^{-1}y$$

$$\frac{\partial L(\theta)}{\partial \theta} = -\frac{2}{n} \sum_{i=1}^N (x^{\{i\}})^T y^{\{i\}} + \frac{2}{n} \sum_{i=1}^N (x^{\{i\}})^T x^{\{i\}} \theta = 0$$

Let's rewrite it as:

$$\frac{\partial L(\theta)}{\partial \theta} = -\frac{2}{n} (x^{\{1\}}, \dots, x^{\{n\}})^T (y^{\{1\}}, \dots, y^{\{n\}}) + \frac{2}{n} (x^{\{1\}}, \dots, x^{\{n\}})^T (x^{\{1\}}, \dots, x^{\{n\}}) \theta = 0$$

Define $X = (x^{\{1\}}, \dots, x^{\{n\}})$ and $y = (y^{\{1\}}, \dots, y^{\{n\}})$

$$\frac{\partial L(\theta)}{\partial \theta} = -\frac{2}{n} X^T y + \frac{2}{n} X^T X \theta = 0$$

$$\Rightarrow \theta = (X^T X)^{-1} X^T y = X^+ y$$

X^+ is the pseudo-inverse of X

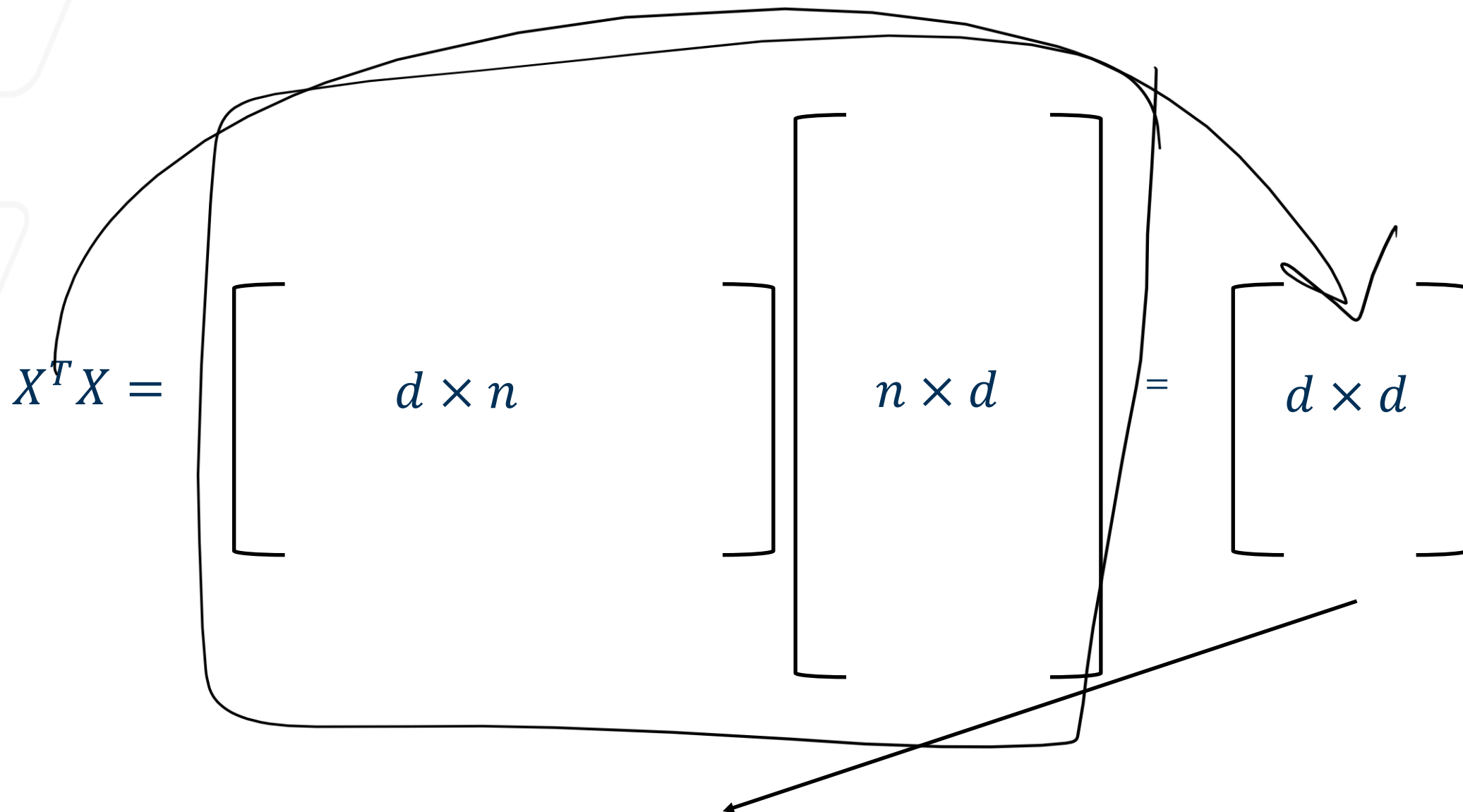
$$X^T X X^+ = X^T$$

$$\underline{\underline{\theta = (X^T X)^{-1} X^T y = X^+ y}}$$

$X_{n \times d}$

$n = \text{instances}$

$d = \text{dimension}$



Not a big matrix because $n \gg d$ This matrix is invertible most of the times.
 If we are *VERY* unlucky and columns of $X^T X$ are not linearly independent (it's not a full rank matrix), then it is not invertible.

Quick Knowledge Check

1. What is the main difference between regression and classification tasks? A. Regression predicts discrete labels, classification predicts continuous outputs. ~~B. Regression predicts continuous values, classification predicts discrete labels.~~ C. Both predict continuous values. D. Both predict categories
2. Which of the following is an example of a regression task? A. Detecting spam emails. ~~B. Predicting a car's fuel efficiency.~~ C. Recognizing handwritten digits. D. Detecting positive vs. negative sentiments in reviews
3. In **backward feature selection**, what happens at each step? A. The feature that least improves performance is added. ~~B. The feature that least reduces performance is removed.~~ C. The feature with the lowest correlation is removed. D. The most important feature is added
4. In the linear regression equation $\hat{y} = w_1x_1 + w_2x_2 + \dots + b$, what does **b** represent? A. The slope of the model. B. The learning rate. ~~C. The bias or intercept term.~~ D. The regularization penalty
5. What is the goal of the least mean square (LMS) method? A. Minimize classification errors. B. Maximize model complexity. ~~C. Minimize the sum of squared residuals between predictions and actual values.~~ D. Minimize correlation between features
6. Why is it important to include a bias term in linear regression? ~~A. To allow the model to fit data that doesn't pass through the origin.~~ B. To normalize input features. C. To regularize the model. D. To reduce computational cost